# Exercise 1: Implementing the Singleton Pattern

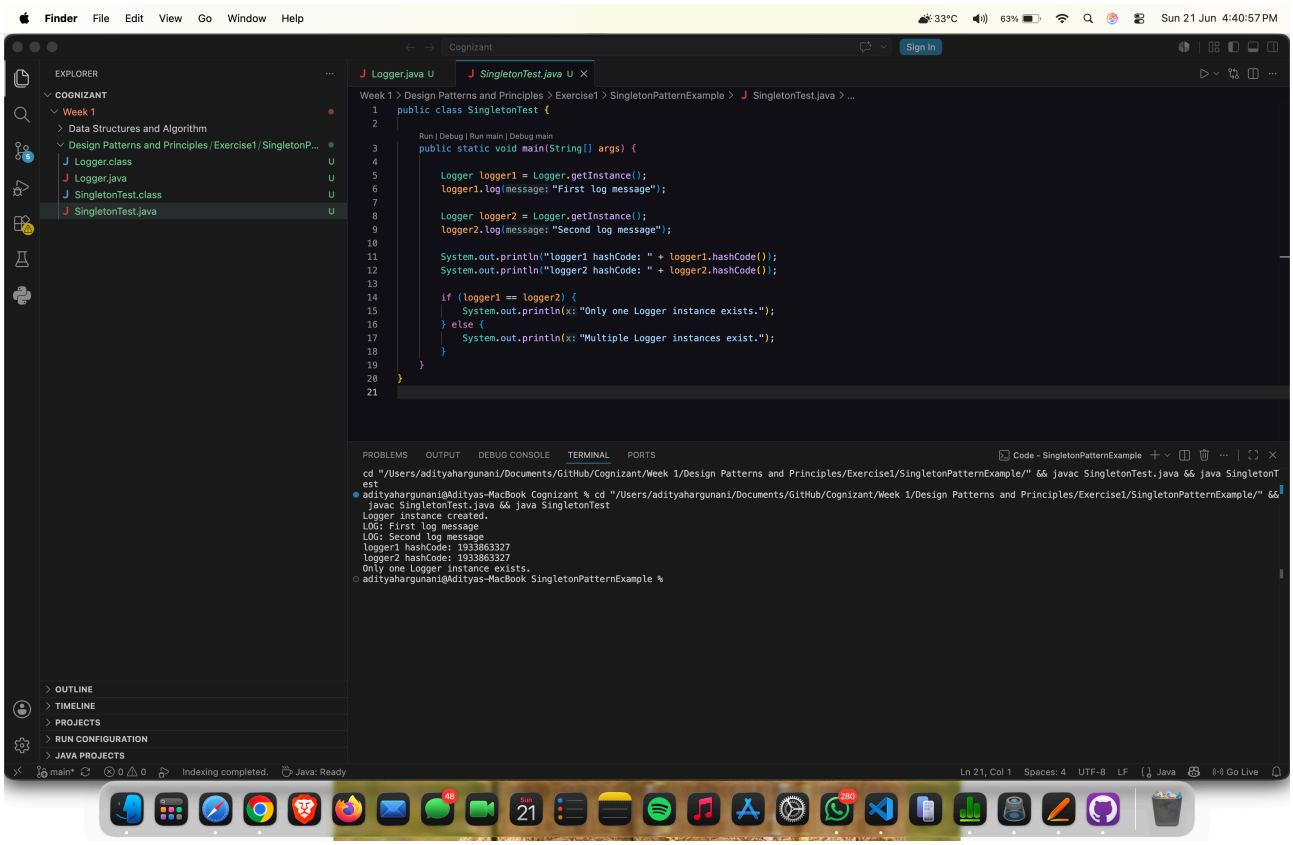
## Logger.java:

```
public class Logger {

    private static Logger instance;

    private Logger() {
        System.out.println("Logger instance created.");
    }

    public static Logger getInstance() {
        if (instance == null) {
            instance = new Logger();
        }
        return instance;
    }

    public void log(String message) {
        System.out.println("LOG: " + message);
    }
}
```

## SingletonTest.java:

```
public class SingletonTest {

    public static void main(String[] args) {

        Logger logger1 = Logger.getInstance();
        logger1.log("First log message");

        Logger logger2 = Logger.getInstance();
        logger2.log("Second log message");

        System.out.println("logger1 hashCode: " + logger1.hashCode());
        System.out.println("logger2 hashCode: " + logger2.hashCode());

        if (logger1 == logger2) {
            System.out.println("Only one Logger instance exists.");
        } else {
            System.out.println("Multiple Logger instances exist.");
        }
    }
}
```

# Output:


Finder File Edit View Go Window Help
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Week 1
Data Structures and Algorithm
Design Patterns and Principles / Exercise1 / SingletonP...
Logger.class
Logger.java
SingletonTest.class
SingletonTest.java
Logger.java U
SingletonTest.java U
Week 1 > Design Patterns and Principles > Exercise1 > SingletonPatternExample > SingletonTest.java > ...
public class SingletonTest {
Run | Debug | Run main | Debug main
public static void main(String[] args) {
Logger logger1 = Logger.getInstance();
logger1.log(message: "First log message");
Logger logger2 = Logger.getInstance();
logger2.log(message: "Second log message");
System.out.println("logger1 hashCode: " + logger1.hashCode());
System.out.println("logger2 hashCode: " + logger2.hashCode());
if (logger1 == logger2) {
System.out.println(x: "Only one Logger instance exists.");
} else {
System.out.println(x: "Multiple Logger instances exist.");
}
}
}
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Code - SingletonPatternExample
cd "/Users/adityahargunani/Documents/GitHub/Cognizant/Week 1/Design Patterns and Principles/Exercise1/SingletonPatternExample/" && javac SingletonTest.java && java SingletonTest
adityahargunani@Adityas-MacBook Cognizant % cd "/Users/adityahargunani/Documents/GitHub/Cognizant/Week 1/Design Patterns and Principles/Exercise1/SingletonPatternExample/" && javac SingletonTest.java && java SingletonTest
Logger instance created.
LOG: First log message
LOG: Second log message
logger1 hashCode: 1933863327
logger2 hashCode: 1933863327
Only one Logger instance exists.
adityahargunani@Adityas-MacBook SingletonPatternExample %
OUTLINE
TIMELINE
PROJECTS
RUN CONFIGURATION
JAVA PROJECTS
main* 0 0 Indexing completed. Java: Ready
Ln 21, Col 1 Spaces: 4 UTF-8 LF Java Go Live